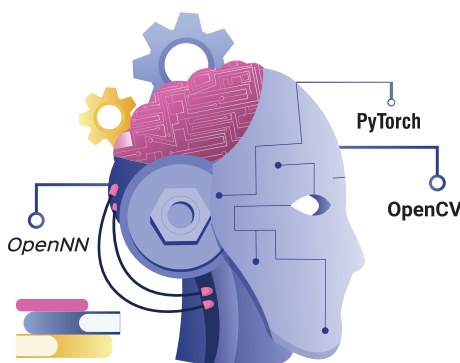


A Few Tools that Help Machines to Learn

Artificial intelligence, machine learning, and deep learning have evolved over decades to help computers perform tasks in the way humans do. This article briefly describes three open source machine learning frameworks – OpenCV, PyTorch and OpenNN – in the context of these technologies.

Artificial intelligence (AI) aims to mimic the functions of the human brain using computers. It uses mathematical and statistical models (e.g., probability) to do so. Machine learning (ML) improves the behaviour of AI models using programming models of brain functions, and storing the results of these models for future predictions.

These technologies are inspired by the human neural network, which is the core storage layer of our brain. The processing efficiency of these neural network based solutions is improved by implementing a distributed model for faster processing, as well as simulation of various brain functions like object detection, analytical ability, etc.



Genetic algorithms improve deep learning models. These have a self-learning ability and are intuitive, hence maximising the applications of AIML (artificial intelligence modelling language) models and reducing the effort in training them. They improve the knowledge base in neural networks.

All these technologies are helping computer based applications to improve their ability to come closer to human-like thinking, while reducing the effort needed to build and train models.

Common features of AIML tools

Many tech giants are investing large sums in deep learning projects. These include Facebook, Intel and Google, to name a few. Google Brain is the AI research team in Google, which was first formed in early 2010 to design deep learning solutions using Google platforms. The first software built by this team was DistBelief. Later, it was instrumental in releasing many popular projects like TensorFlow, Google Analytics and Google Translate.

There are many popular machine learning frameworks available in the market, many of which are open source. However, we cannot blindly select a machine learning tool or framework based on convenience. We need to look at the problem scenario, data requirements, data volume to process, training data availability, trainability, algorithm support, visualisation and reporting facility, capability to auto-build the knowledge or training model, processing speed required (CPU/GPU), to name a few, before we make a choice.

In machine learning, building the model is a time consuming task as a lot of training is needed to get higher accuracy in processing the model for

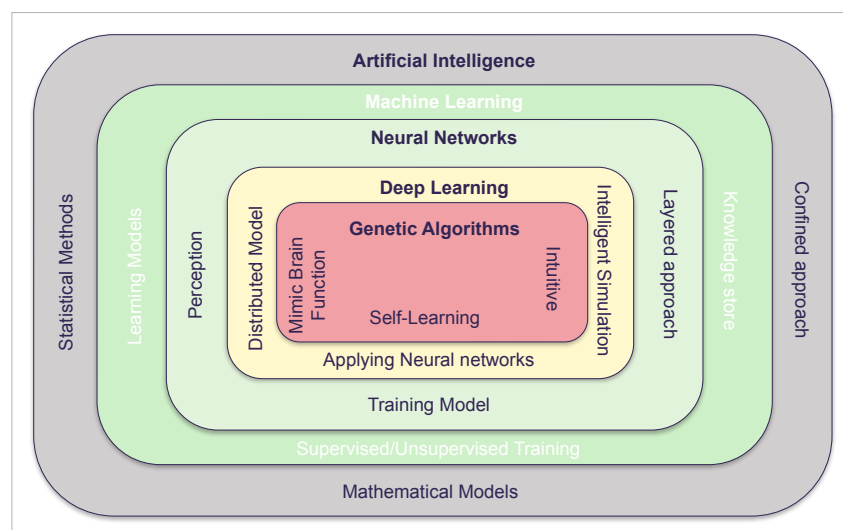


Figure 1: AIML model functions